

EXP19: Prolog Program for Student-Teacher-Subject-Code

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Aim:

To develop a Prolog program that maintains a database relating Students, Teachers, Subjects and Subject Codes, and to query the details for each student.

Algorithm:

- Step 1: Start the program.
- Step 2: Create facts representing teachers, subjects, and subject codes.
- Step 3: Create facts representing students and the subjects they study.
- Step 4: Select a student record from the database.
- Step 5: Identify the subject studied by that student.
- Step 6: Match the subject with the corresponding teacher and subject code.
- Step 7: Display the student name, subject, teacher name, and subject code.
- Step 8: Repeat the process for all student records.
- Step 9: Stop the program.

Source Code:

% Program for Student-Teacher-Subject-Code Database

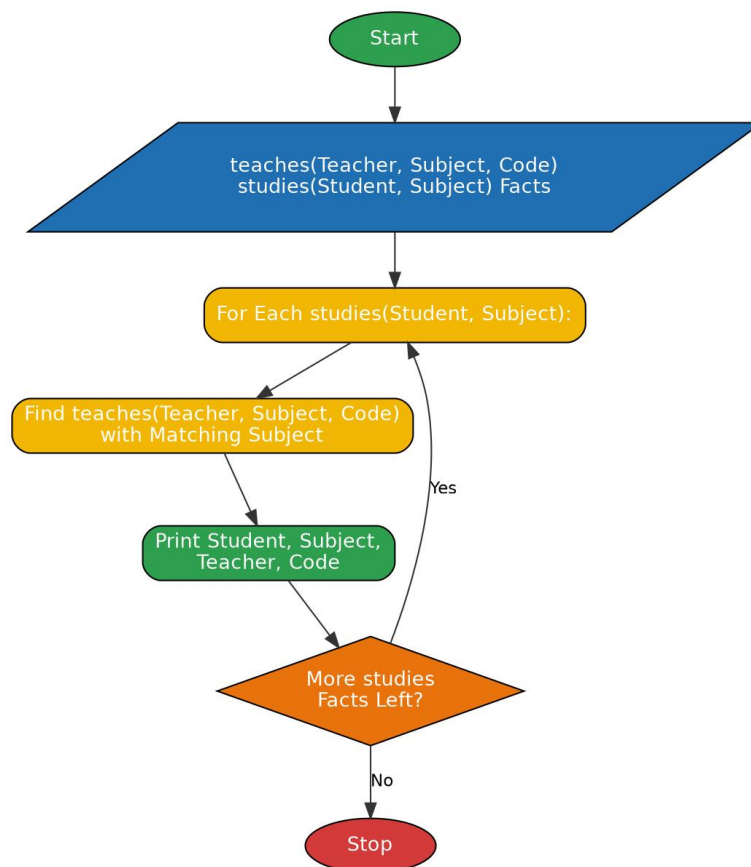
```
teaches(mr_raj, maths, cs101).  
teaches(mrs_priya, physics, cs102).  
teaches(mr_kumar, chemistry, cs103).
```

```
studies(ram, maths).  
studies(sita, physics).  
studies(geeta, chemistry).  
studies(hari, maths).
```

show_details :-

```
    forall(studies(Student, Subject),  
        ( teaches(Teacher, Subject, Code),  
          format("~w studies ~w, taught by ~w (Code: ~w)~n", [Student, Subject, Teacher,  
Code])  
        )  
    ).
```

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [teacherdb].
true.

?- show_details.
ram studies maths, taught by mr_raj (Code: cs101)
sita studies physics, taught by mrs_priya (Code: cs102)
geeta studies chemistry, taught by mr_kumar (Code: cs103)
hari studies maths, taught by mr_raj (Code: cs101)
true.

?-
```

Result:

Thus the Prolog program for the Student-Teacher-Subject-Code database was written, executed successfully and the output was verified.